



Indian Institute of Technology Bhubaneswar
Department of Computer Science and Engineering, SECS
End-Semester (Spring) Examination – 2026

Subject Name: Advanced Computer Networks (ACN)		Subject Code: CS6L048
Date: 29/04/2026	Duration: 3 hours	Full Marks: 120
• Answers without sufficient justification will not be accepted. • Answer all the questions (as instructed).		

(Q1) In a network system, packet arrivals follow a **Poisson process** with arrival rate: $\lambda = 2$ packets per ms. Consider a time interval of length $T = 1$ ms.

- (a) Compute the **probability that exactly 2 packets arrive** during the interval T
- (b) Compute the **probability that no packet arrives** during the interval T
- (c) Given that **at least one packet has arrived**, compute the **conditional probability that exactly one packet arrived**.
- (d) Compute the **expected number of arrivals** in an interval of **3 ms**, and also find the **probability that no arrivals occur in this 3 ms interval**.

[Marks 5+ 5+ 5+ 5 =20]

Answer:

In a Poisson process, the probability of observing exactly k events in a time interval T is given by the formula: $P(N=k) = \frac{(\lambda T)^k}{k!} e^{(-\lambda T)}$

$$\lambda = 2 \text{ packets/ms}$$

$$T = 1 \text{ ms}$$

- (a) **Probability that exactly 2 packets arrive** during the interval T

$$P(N=2) = \frac{(2 \times 1)^2}{2!} e^{(-2 \times 1)} = \frac{2^2}{2!} e^{-2} = 0.2707$$

(b) **Probability that no packet arrive during the interval T**

$$P(N=0) = \frac{(2 \times 1)^0}{0!} e^{(-2 \times 1)} = e^{-2} = 0.1353$$

(c) **Probability that exactly one arrived (Given that at least one packet arrived)**

$$P(N=1|N \geq 1) = \frac{P(N=1)}{P(N \geq 1)}$$

$$\text{Probability of exactly 1 arrival, } P(N=1) = \frac{2^1}{1!} e^{-2} = 2e^{-2} = 0.2707$$

$$\text{Probability of at least 1 arrival, } P(N \geq 1) = 1 - P(N=0) = 1 - e^{-2} = 0.8647$$

$$P(N=1|N \geq 1) = \frac{0.2707}{0.8647} = 0.3130$$

(d) Compute the **expected number of arrivals** in an interval of **3 ms**, and also find the **probability that no arrivals occur in this 3 ms interval**.

Expected number of arrivals in 3 ms

$$E[N] = \lambda t = 2 \times 3 = 6$$

For a new interval $T = 3$ ms

$$P(N=0) = \frac{(2 \times 3)^0}{0!} e^{(-2 \times 3)} = \frac{(6)^0}{0!} e^{-6} = 0.00248$$

(Q2) Consider a system modelled as a **Birth–Death Markov process**, where: Arrival rate (birth rate): $\lambda = 2$ customers/sec, Service rate (death rate): $\mu = 4$ customers/sec. Let p_k denote the **steady-state probability** that there are k customers in the system. Assume: *Infinite buffer (system can hold any number of customers)*, and the system has reached **steady state**.

- (a) Compute & explain the key factor (ρ) based on which we can determine system stability
- (b) Find steady-state probabilities: p_0 (no customer) , p_1 (one customer) and p_2 (two customers)
- (c) Find the **probability that there are at least 2 customers in the system**, i.e., $P(N \geq 2)$
- (d) Compute the **expected number of customers in the system**, $E[N]$

[Marks 5 + 5 + 5 + 5 =20]

Answer:

For a Birth–Death process with an infinite buffer, this is the standard M/M/1 steady-state model.

(Given): $\lambda=2$, $\mu=4$

(a) Compute & explain the key factor (ρ) - To determine system stability

The key factor for determining stability is the **traffic intensity** (also called the utilization factor), denoted by ρ . It is the ratio of the arrival rate to the service rate:

$$\rho = \frac{\lambda}{\mu} = \frac{2}{4} = 0.5$$

Explanation: Condition (from queueing theory): $\rho < 1 \Rightarrow$ steady state exists

- The queue will not grow indefinitely because the server is fast enough to handle the incoming load.

- Server is faster → queue clears → system reaches equilibrium

Since $\rho=0.5<1$, the system is **stable**.

(b) Find steady-state probabilities: p_0 (no customer) , p_1 (one customer) and p_2 (two customers)

From Birth–Death steady-state relation :

$$P[\text{ k customers in system}], p_k = \rho^k (1 - \rho)$$

$$\text{Probability of 0 customers } (p_0): p_0 = (1 - \rho) = 1 - 0.5 = 0.5$$

$$\text{Probability of 1 customer } (p_1): p_1 = \rho^1 p_0 = 0.5 \times 0.5 = 0.25$$

$$\text{Probability of 2 customers } (p_2): p_2 = \rho^2 p_0 = 0.25 \times 0.5 = 0.125$$

(c) Find the **probability that there are at least 2 customers in the system**, i.e., $P(N \geq 2)$

$$P(N \geq 2) = 1 - (p_0 + p_1) = 1 - (0.5 + 0.25) = 0.25$$

(d) Compute the **expected number of customers in the system**, $E[N]$

$$N(M/M/1) = \frac{\rho}{1 - \rho} = \frac{0.5}{1 - 0.5} = 1$$

On average, there is 1 customer in the system at any given time.

(Q3) Consider a **Pure ALOHA system** where: Packet transmission time, **T = 1 ms**, Total offered load, **G = 0.5 attempts per packet time**. Packet arrivals follow a **Poisson process**. Nodes transmit immediately upon arrival (no sensing). A packet is successful only if **no other transmission occurs during its vulnerable period**

(a) Determine the **vulnerable period** for Pure ALOHA and explain why it has this duration

(b) Compute the **probability that a packet is successfully transmitted**

(c) Compute the **throughput (S)** of the system.

(d) Let the offered load doubles (**G = 1**). Find new throughput. Explain details.

[Marks 5 + 5 + 5 + 5=20]

Answer:

(a) Determine the **vulnerable period** for Pure ALOHA and explain why it has this duration

For **Pure ALOHA**, a packet can start at any time. So, if a packet starts at time t , then any other packet that starts in $(t-1, t+1]$ will collide with it.

So the **vulnerable period is $2T$** .

Since $T = 1$ ms, Vulnerable period = $2T = 2$ ms

(b) Compute the **probability that a packet is successfully transmitted**

$$P_{\text{success}} = P[0 \text{ arrivals in } 2 \text{ time units}] = e^{-2G}$$

Given: $G = 0.5$

$$P_{\text{success}} = e^{-2(0.5)} = e^{-1} = 0.3679$$

(c) Compute the **throughput (S)** of the system

Given: $G = 0.5$

$$S = Ge^{-2G} = 0.5 \times e^{-1} = 0.5 \times 0.3679 = \mathbf{0.1839}$$

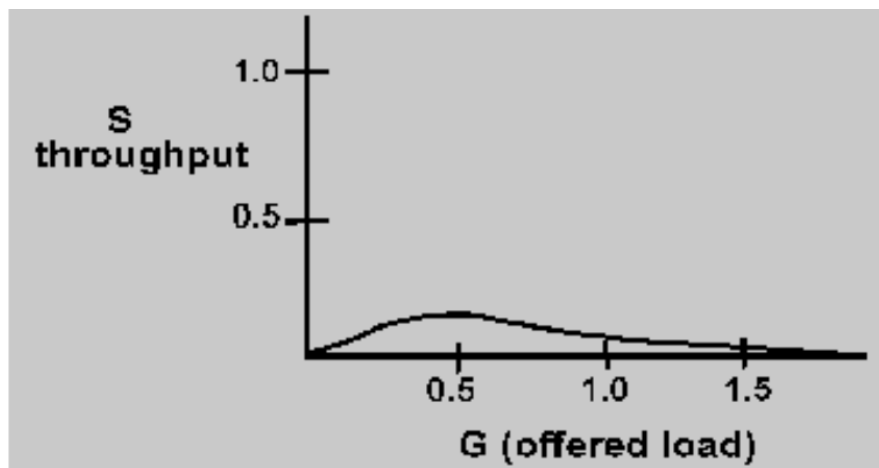
(d) Let the offered load doubles ($G = 1$). Find new throughput. Explain details

Given: new $G = 1$

$$S_{new} = Ge^{-2G} = 1 \times e^{-2(1)} = e^{-2} = \mathbf{0.1353}$$

When the offered load increases beyond $G=0.5$, the number of transmission attempts increases, but the probability of collision increases even faster due to the large vulnerable period $2T$.

Hence, the success probability e^{-2G} decreases exponentially, causing the throughput to drop. Thus the throughput drops from 0.1839 to 0.1353, even though offered load increased.



(Q4) Answer either Part 1 or Part 2 (Any one). Clearly mention which part you are answering.

Part 1: Consider an **unslotted 0-persistent CSMA system** with the following parameters: Packet transmission time $T = 1 \text{ ms}$, Maximum propagation delay $\tau = 0.25 \text{ ms}$, Offered load, $g = 2 \text{ packets/ms}$. Assume the following: *Poisson arrival process, All packets have equal length, A transmission is successful **only if no other packet arrives during the vulnerable period***. Consider the following approximation in this calculation: $B \approx T + \tau$ (B is the average duration of the busy period).

- (a) Compute the **probability of successful transmission**
- (b) Compute the **average useful time (U)**
- (c) Compute the **average idle time (I)**
- (d) Compute the **throughput (S)** using: $S = U/(B+I)$

[Marks 5 + 5 + 5 + 5 = 20]

Answer

Using the CSMA 0-persistent notes, a transmission is successful if no other packet starts within the vulnerable interval τ after the first one begins, so the success probability is $e^{-g\tau}$

Given:

$$T = 1 \text{ ms}$$

$$\tau = 0.25 \text{ ms}$$

$$g = 2 \text{ packets/ms}$$

(a) Compute the **probability of successful transmission**: Using the CSMA 0-persistent notes, a transmission is successful if no other packet starts within the vulnerable interval τ after the first one begins, so the success probability is $e^{-g\tau}$

$$P_{\text{success}} = e^{-g\tau} = e^{-2 \times 0.25} = e^{-0.5} = 0.6065$$

(b) Compute the **average useful time (U)**: Average useful time represents the portion of a cycle spent transmitting successful data. It is the product of the transmission time and the probability of success:

$$U = T \cdot e^{-g\tau} = 1 \times 0.6065 = 0.6065 \text{ ms}$$

(c) Compute the **average idle time (I)**: The idle time is the average duration the channel remains empty between busy periods. Since packet arrivals follow a Poisson process with rate g , the time between arrivals is exponentially distributed with mean $1/g$:

$$I = \frac{1}{g} = \frac{1}{2} = 0.5 \text{ ms}$$

(d) Compute the **throughput (S)** using: $S = U/(B+I)$

Using the given approximation: $B \approx T + \tau = 1 + 0.25 \text{ ms} = 1.25 \text{ ms}$

$$S = \frac{U}{B+I} = \frac{0.6065}{1.25 + 0.5} = \frac{0.6065}{1.75} = 0.3466$$

Under these conditions, the channel utilization is approximately 34.66%.

Part 2: Consider an **unslotted 0-persistent CSMA network** with: Packet transmission time **$T = 1 \text{ ms}$** , Propagation delay: **$\tau = 0.2 \text{ ms}$** , Offered load: **$g = 1.5 \text{ packets/ms}$** . Assume that the system alternates between **idle period and busy period** cycles. A busy period is **useful** if exactly one transmission occurs, and is **not useful** if collision occurs.

Should we consider the following approximation in this calculation: **$B \approx T + \tau$**
(**B** is the average duration of the busy period).

- (a) What is the **probability that a busy period is useful**?
- (b) What is the **expected fraction of time wasted due to collisions**?
- (c) Compute the **throughput (S)**
- (d) If propagation delay doubles (i.e., τ becomes 2τ), **how much does throughput degrade (in percentage) ?**

[Marks 5+5+5+5 = 20]

Answer

Using the CSMA 0-persistent notes, a transmission is successful if no other packet starts within the vulnerable interval τ after the first one begins, so the success probability is $e^{-g\tau}$

Given:

$$T = 1 \text{ ms}$$

$$\tau = 0.2 \text{ ms}$$

$$g = 1.5 \text{ packets/ms}$$

(a) What is the **probability that a busy period is useful**.

A busy period is useful $\Leftrightarrow$ exactly one transmission $\Leftrightarrow$ no arrivals in vulnerable period τ

$$P_{\text{useful}} (= P_{\text{success}}) = e^{-g\tau} = e^{-1.5 \times 0.2} = e^{-0.3} = \mathbf{0.7408}$$

(b) What is the expected fraction of time wasted due to collisions?

Using the approximation:

$$B \approx T + \tau = 1 + 0.2 = 1.2 \text{ ms}$$

The probability that a busy period results in a collision is:

$$P_{\text{collision}} = 1 - e^{-g\tau} = 1 - e^{-1.5 \times 0.2} = 0.2592$$

The average idle period is:

$$I = \frac{1}{g} = \frac{1}{2} = 0.5 \text{ ms}$$

Now, the fraction of time wasted due to collisions is:

$$\text{Fraction of time wasted} = \frac{(P_{\text{collision}} \times B)}{(I + B)}$$

Substituting the values:

$$\text{Fraction of time wasted} = \frac{(0.2592 \times 1.2)}{(0.6667 + 1.2)}$$

$$\text{Fraction of time wasted} = 0.1666$$

Therefore,

$$\text{Fraction of time wasted} \approx 16.7\%$$

Final Answer: The expected fraction of time wasted due to collisions is approximately 16.7%.

(c) Compute the throughput (S)

$$S = \frac{U}{B+I} = \frac{0.7408}{1.2 + 0.6667} = \frac{0.7408}{1.8667} = 0.397$$

$$\text{Useful time, } U = T \cdot P_{\text{success}} = 1 \times 0.7408 = 0.7408 \text{ ms}$$

$$\text{Busy period, } B \approx T + \tau = 1 + 0.2 \text{ ms} = 1.2 \text{ ms}$$

$$(\text{ OR } B = T + 2\tau - \frac{1-e^{-g\tau}}{g} = 1.2272 \text{ ms})$$

$$\text{Idle Time, } I = \frac{1}{g} = \frac{1}{1.5} = 0.6667 \text{ ms}$$

OR

Using the standard formula for unslotted non-persistent CSMA:

$$S = \frac{Ge^{-aG}}{G(1+2a) + e^{-aG}}$$

$$S = \frac{1.5 \cdot e^{-0.3}}{1.5(1+0.4) + e^{-0.3}} = \frac{1.5 \cdot 0.7408}{2.1 + 0.7408} = \frac{1.1112}{2.8408}$$

$$S \approx \mathbf{0.3912}$$

(d) If propagation delay doubles (i.e., τ becomes 2τ), how much does throughput degrade (in percentage) ?

$$\tau = 0.2 \text{ ms}$$

If propagation delay doubles

$$T_{\text{new}} = 0.4 \text{ ms}$$

$$P_{\text{success}} = e^{-g\tau} = e^{-1.5 \times 0.4} = e^{-0.6} = \mathbf{0.5488}$$

$$\text{New Useful time, } U_{\text{new}} = T \cdot P_{\text{success}} = 1 \times 0.5488 = 0.5488 \text{ ms}$$

$$\text{New Busy period, } B_{\text{new}} \approx T + \tau = 1 + 0.4 \text{ ms} = 1.4 \text{ ms}$$

$$\text{New Idle Time, } I_{\text{new}} = \frac{1}{g} = \frac{1}{1.5} = 0.6667 \text{ ms}$$

$$S_{\text{new}} = \frac{U_{\text{new}}}{B_{\text{new}} + I_{\text{new}}} = \frac{0.5488}{1.4 + 0.6667} = \frac{0.5488}{2.0667} = \mathbf{0.2655}$$

$$\text{Percentage degradation, \% decrease} = \frac{S_{\text{old}} - S_{\text{new}}}{S_{\text{old}}} \times 100$$

$$\% \text{ decrease} = \frac{0.397 - 0.2655}{0.397} \times 100 = \mathbf{33.1\%}$$

(Q5) Consider an **unslotted 1-persistent CSMA system** with: Packet transmission time: $T = 1 \text{ ms}$, Propagation delay: $\tau = 0.2 \text{ ms}$, Arrival process: Poisson with rate $g = 1 \text{ packet/ms}$. At the **start of a transmission period**, exactly **one node begins transmission** (i.e., the system is in **state 1 / Type-1 period**). During the **vulnerable period (τ)**: Other nodes that have packets, **sense busy channel and will transmit immediately after current transmission ends** (due to 1-persistence behavior).

- (a) What is the **probability that this Type-1 period remains collision-free**?
- (b) What is the **expected number of packets that will attempt transmission immediately after the current packet finishes**, given that arrivals occurred during the vulnerable period?
- (c) Based on your result, determine whether the **next period is more likely to be: Type 0** (idle), **Type 1** (single transmission), **Type 2** (collision) (quantitatively, derive the probabilities).
- (d) Explain how this behavior differs fundamentally from **0-persistent CSMA**, in terms of **queue build-up and collision clustering**.

[Marks 5 + 5 + 5 + 5 = 20]

Answer:

System Parameters:

- Packet transmission time (T): 1 ms
- Propagation delay (τ): 0.2 ms
- Arrival process (g): Poisson with rate 1 packet/ms
- Expected arrivals in time t (N_t): Poisson($g * t$)

Scenario 1: Standard CSMA-1p Model [Based on what we learnt in class, not considering the simplification suggested in the question]

Conceptual Assumption: In real-world networks, propagation delay creates a "vulnerable period" (τ) at the very start of a transmission. During this 0.2 ms window, a newly transmitting node's signal has not yet reached the other nodes. Therefore, if another node decides to transmit during this brief window, it will falsely sense the channel as "idle" and begin transmitting, causing an unavoidable collision during the current transmission period.

(a) What is the probability that this Type-1 period remains collision-free?

Probability that this Type-1 period remains collision-free A Type-1 period is successful only if there are zero arrivals during the vulnerable period τ .

$$P(\text{collision-free}) = P(N_\tau = 0) = e^{-g\tau} = e^{-1(0.2)} \approx 0.8187 \quad \text{Answer: 0.8187}$$

(b) What is the expected number of packets that will attempt transmission immediately after the current packet finishes, given that arrivals occurred during the vulnerable period?

If arrivals occurred during τ , a collision happens. The busy period extends slightly because the colliding packets start up to τ ms later. Let y be the time of the last arrival in τ . The channel remains busy until this last packet finishes, making the total busy time $y + T + \tau$. The accumulation time (where nodes sense busy) is $(y + T + \tau) - \tau = y + T$.

The unconditional expected value of y is:

$$E[y] = \tau - \frac{1 - e^{-g\tau}}{g} = 0.2 - \frac{1 - e^{-1(0.2)}}{1} = 0.0187$$

Conditioned on an arrival actually occurring ($y > 0$), we calculate:

$$E[y | y > 0] = \frac{E[y]}{P[y > 0]} = \frac{E[y]}{1 - P[\text{collision}]} = \frac{0.0187}{0.1813} = 0.103 \text{ ms}$$

Expected accumulated packets: $g[T + E[y | y > 0]] = 1[1 + 0.103] = 1.103$ packets

Answer: 1.1033 packets.

In this standard model, these packets do not wait; they transmit and collide during the current transmission period.

(c) Next period probabilities Using the exact slide-based Markov analysis formulas, where the vulnerable-period arrivals affect the current period length:

- Type 0 (Idle): $P_{10} = [1 + g\tau]e^{-g(T+\tau)} \approx \mathbf{0.3614}$
- Type 1 (Single): $P_{11} = [T + g\tau[T + \frac{\tau}{2}]]ge^{-g(T+\tau)} \approx \mathbf{0.3675}$
- Type 2 (Collision): $P_{12} = 1 - p_{10} - p_{11} \approx \mathbf{0.2711}$

Scenario 2: Ideal Sensing Model [Considering the simplification suggested in the question]

Assumption: Nodes have perfect, instantaneous sensing. They can sense the channel is busy even during the vulnerable period. Because they are 1-persistent, they do not drop their packets; instead, they wait and queue up to transmit exactly at the millisecond the current transmission ends.

(a) Probability that this Type-1 period remains collision-free In this ideal sensing scenario, arriving nodes perfectly detect the ongoing transmission from the very start. Because they sense it is busy, they wait rather than transmit. Since no node will ever interrupt an ongoing transmission they can hear, there is zero chance of interference during the current packet.

Answer: Probability = 1

(b) Concept: Because nodes wait instead of colliding, we must calculate the total number of packets that accumulate in the "waiting pool" during the 1 ms transmission. We are calculating this GIVEN the strict condition that at least one packet arrived during the vulnerable period τ .

Derivation: We split the transmission time into two phases and add them together.

- Phase 1 (The vulnerable period, τ): We already know at least 1 packet arrived here. Using the conditional expectation formula from Scenario 1: **1.1033 packets.**
- Phase 2 (The remaining time, $T - \tau$): Because of the "memoryless" property of a Poisson process, whatever happened in the first 0.2 ms has absolutely no impact on the remaining 0.8 ms of the transmission. We calculate this normally.
- Expected arrivals = Remaining Time = $1(0.8) = \mathbf{0.8 \text{ packets}}$
- Total expected waiting pool = $1.1033 + 0.8 = \mathbf{1.9033}$.

Answer: 1.9033 packets

(c) Next period probabilities, we simply calculate based on the total packets ($N_{T+\tau}$) arriving during the entire transmission time and propagation delay). So all the arrivals follow Poisson arrival process

Since the accumulated packets follow a Poisson distribution, we use the standard Poisson probability formula: $P(N_{T+\tau} = k) = \frac{(g(T+\tau))^k}{k!} e^{-g(T+\tau)}$

- Type 0 (Idle): $P(N_{T+\tau} = 0) = \frac{(1*(1+0.2))^0}{0!} e^{-1*(1+0.2)} \approx 0.3010$

Answer: 0.3010

- Type 1 (Single): $P(N_{T+\tau} = 1) = \frac{(1*(1+0.2))^1}{1!} e^{-1*(1+0.2)} \approx 0.3612$

Answer: 0.3612

- Type 2 (Collision):

$$P(N_T \geq 2) = 1 - P(\text{Type 0}) - P(\text{Type 1}) = 1 - 0.3010 - 0.3612 = 0.3378$$

Answer: 0.3378

(d) Explain how this behavior differs fundamentally from **0-persistent CSMA**, in terms of **queue build-up and collision clustering**.

In 0-persistent CSMA, a node sensing a busy channel does not stick to the current busy period; it waits a random backoff and retries later. That spreads attempts out and reduces synchronized pile-ups. In 1-persistent CSMA, nodes keep listening and then transmit immediately when the channel becomes free, so all arrivals during the vulnerable period tend to bunch together at the end of the current transmission. This creates collision clustering and stronger queue build-up inside a busy period. The slides explicitly describe that in 1-persistent CSMA, nodes that sense busy “transmit immediately after the current transmission ends,” and that Type 2 periods arise when multiple packets accumulate during the vulnerable period.

Feature	0-persistent CSMA	1-persistent CSMA
Reaction to Busy Channel	Backs off (waits random time).	Persists (waits at the "gate").
Collision Clustering	Low. Arrivals are spread out over time.	High. All packets arriving during a busy period collide at once at the end.
Throughput at Low Load	Slightly lower (due to unnecessary idle time).	Higher. No delay in starting the next transmission.
Throughput at High Load	Stable.	Collapses. High loads lead to massive "pile-up" collisions.

(Q6) Answer either Part 1 or Part 2 (Any one). Clearly mention which part you are answering.

Part 1:

Consider two TCP flows sharing a bottleneck link of capacity **C = 100 units**. Their rates evolve according to the following control laws: **Additive Increase (no congestion):** $x_i(t+1) = x_i(t) + 10$, and **Multiplicative Decrease (on congestion):** $x_i(t+1) = 0.5 \cdot x_i(t)$. Initial rates are: $x_1 = 20$, $x_2 = 60$. Congestion occurs whenever $x_1 + x_2 > 100$. The system alternates between increase phases and decrease events.

- (a)** After one **increase step**, compute the new rates and check whether congestion occurs
- (b)** If congestion occurs, apply the **decrease rule** and compute the updated rates
- (c)** Compute the **fairness index** before and after the decrease using: $F = \frac{(x_1 + x_2)^2}{2(x_1^2 + x_2^2)}$

(d)Based on your results, determine whether the system is moving toward:
Fairness ($x_1 \approx x_2$) and/or **Efficiency** ($x_1 + x_2 \approx C$) [Numerically with graphical interpretation].

[Marks5+5+ 5 + 5 = 20]

Given
 $C=100$
 $x_1=20$
 $x_2=60$

Increase rule: $x_i(t+1)=x_i(t)+10$

Decrease rule: $x_i(t+1)=0.5x_i(t)$

(a)After one **increase step**, compute the new rates and check whether congestion occurs

$$x_1 = 20 + 10 = 30$$

$$x_2 = 60 + 10 = 70$$

$$\text{Total Rate: } x_1 + x_2 = 30 + 70 = 100$$

$$x_1 + x_2 = 100 > 100$$

No congestion occurs; Since $x_1 + x_2 \leq 100$, congestion does **not** technically occur at this exact step. However, any further increase would trigger it. Assuming the system takes one more step or reacts to the boundary:

(b)If congestion occurs, apply the **decrease rule** and compute the updated rates

Next increase:

$$x_1=40, x_2=80$$

$$x_1 + x_2 = 120 > 100 \Rightarrow \text{congestion}$$

Apply multiplicative decrease:

$$x_1 = 0.5 \times 40 = 20,$$

$$x_2 = 0.5 \times 80 = 40$$

(c) Compute the **fairness index** before and after the decrease using: $F = \frac{(x_1 + x_2)^2}{2(x_1^2 + x_2^2)}$

$$\text{Before decrease (40, 80): } F = \frac{(120)^2}{2(40^2 + 80^2)} = \frac{14400}{2(1600 + 6400)} = 0.9$$

$$\text{After decrease (20, 40): } F = \frac{(60)^2}{2(20^2 + 40^2)} = \frac{3600}{2(400 + 1600)} = 0.9$$

Fairness unchanged after decrease

(d) Based on your results, determine whether the system is moving toward:
Fairness ($x_1 \approx x_2$) and/or **Efficiency** ($x_1 + x_2 \approx C$) [Numerically with graphical interpretation].

After decrease:

- Ratio remains $x_1:x_2=1:2$
- So fairness **does not improve immediately**

However:

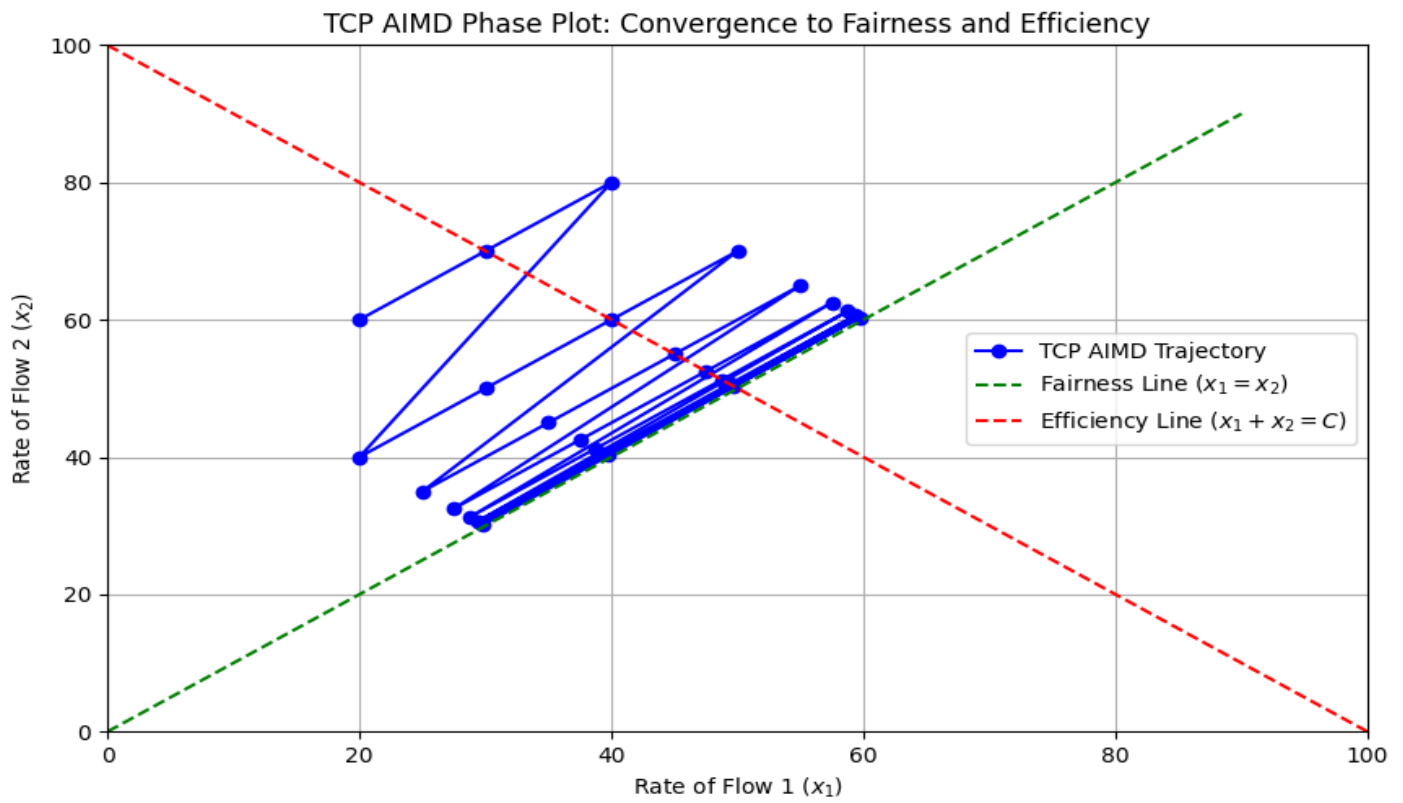
- During next increase:
 - Both increase equally $\rightarrow$ gap reduces

Fairness ($x_1 \approx x_2$): During additive increase, the absolute difference $|x_1 - x_2|$ remains constant, but the *relative* difference decreases. Over multiple cycles, the state vector is pushed toward the 45 deg "Fairness Line."

Efficiency ($x_1 + x_2 \approx C$): The system constantly probes for bandwidth until it hits the "Efficiency Line" ($x_1 + x_2 = 100$), then backs off and repeats.

Therefore:

- Additive increase → improves fairness
- Multiplicative decrease → preserves ratio



Part 2:

Consider a TCP connection with: Initial congestion window: **CWND = 1 MSS**, Threshold: **SSThresh = 16 MSS**, RTT is constant, No delayed ACK (1 ACK per segment/packet). In summary, the behavior rules are as follows: **Slow Start**: CWND doubles every RTT, **Congestion Avoidance**: CWND increases by **1 MSS per RTT**, **On packet loss (timeout)**: $SSThresh = CWND/2$, and $CWND = 1$. Assume the following sequence of events: (1) Transmission starts at $t = 0$ (2) No loss occurs until **CWND reaches 20 MSS** (3) At that point, a **timeout occurs** (4) After recovery, the connection continues normally.

- (a) Determine the value of **CWND at the end of each RTT** until it first reaches **20 MSS** (provide in the form of a table)
- (b) At the moment of packet loss ($CWND = 20 MSS$), compute the new values of: **SSThresh, CWND (after timeout)**
- (c) After the timeout, how many RTTs are required for CWND to again reach the **new SSThresh**?
- (d) Compute the **total number of segments transmitted** from the beginning until CWND again reaches the new SSThresh after recovery

[Marks 5+5+5+5 = 20]

a) CWND at the End of Each RTT

During the **Slow Start** phase, CWND doubles every RTT until it hits the SSThresh (16 MSS). Once CWND reaches 16 MSS, the **Congestion Avoidance** phase begins, where CWND increases by 1 MSS per RTT.

Transmission starts with CWND = 1 MSS. The progression is as follows:

RTT	Phase	CWND at Start of RTT	Segments Sent	CWND at End of RTT
1	Slow Start	1 MSS	1	2 MSS
2	Slow Start	2 MSS	2	4 MSS
3	Slow Start	4 MSS	4	8 MSS
4	Slow Start	8 MSS	8	16 MSS (Reaches SSThresh)
5	Congestion Avoidance	16 MSS	16	17 MSS
6	Congestion Avoidance	17 MSS	17	18 MSS
7	Congestion Avoidance	18 MSS	18	19 MSS
8	Congestion Avoidance	19 MSS	19	20 MSS

At the end of RTT 8, CWND reaches 20 MSS. In RTT 9, a full window of 20 MSS is sent, and a timeout occurs.

b) At the Moment of Packet Loss (Timeout)

When a timeout occurs at $CWND = 20 \text{ MSS}$, TCP applies its penalty rules:

- **New SSThresh** is set to half of the current CWND: $SSThresh = 20 / 2 = 10 \text{ MSS}$
- **New CWND** is dropped back to 1 MSS to restart the Slow Start phase:
 $CWND = 1 \text{ MSS}$

c) RTTs Required to Reach the New SSThresh

After the timeout, the connection is back in the Slow Start phase trying to reach the new SSThresh of **10 MSS**.

RTT	Phase	CWND at Start of RTT	Segments Sent	CWND at End of RTT
1(POST TIMEOUT)	Slow Start	1 MSS	1	2 MSS
2	Slow Start	2 MSS	2	4 MSS
3	Slow Start	4 MSS	4	8 MSS
4	Slow Start	8 MSS	8	16 MSS (Reaches SSThresh)

It takes exactly **4 RTTs** after the timeout for CWND to reach the new SSThresh limit.

d) Total Segments Transmitted

To find the total number of segments transmitted from the very beginning until CWND hits the new SStresh (10 MSS) after recovery, we sum the segments sent in all previous steps:

Before timeout (RTT 1 to RTT 9):

- Slow Start: $1 + 2 + 4 + 8 = 15$ segments
- Congestion Avoidance: $16 + 17 + 18 + 19 = 70$ segments
- RTT 9 (the loss event): 20 segments were sent before the timeout was detected.
- *Total before timeout = $15 + 70 + 20 = 105$ segments.*

After timeout (Recovery to SStresh):

- RTT 1 to RTT 4: $1 + 2 + 4 + 8 = 15$ segments sent.

Total Segments Transmitted:

105 (before loss) + 15 (during recovery) = **120segments.**